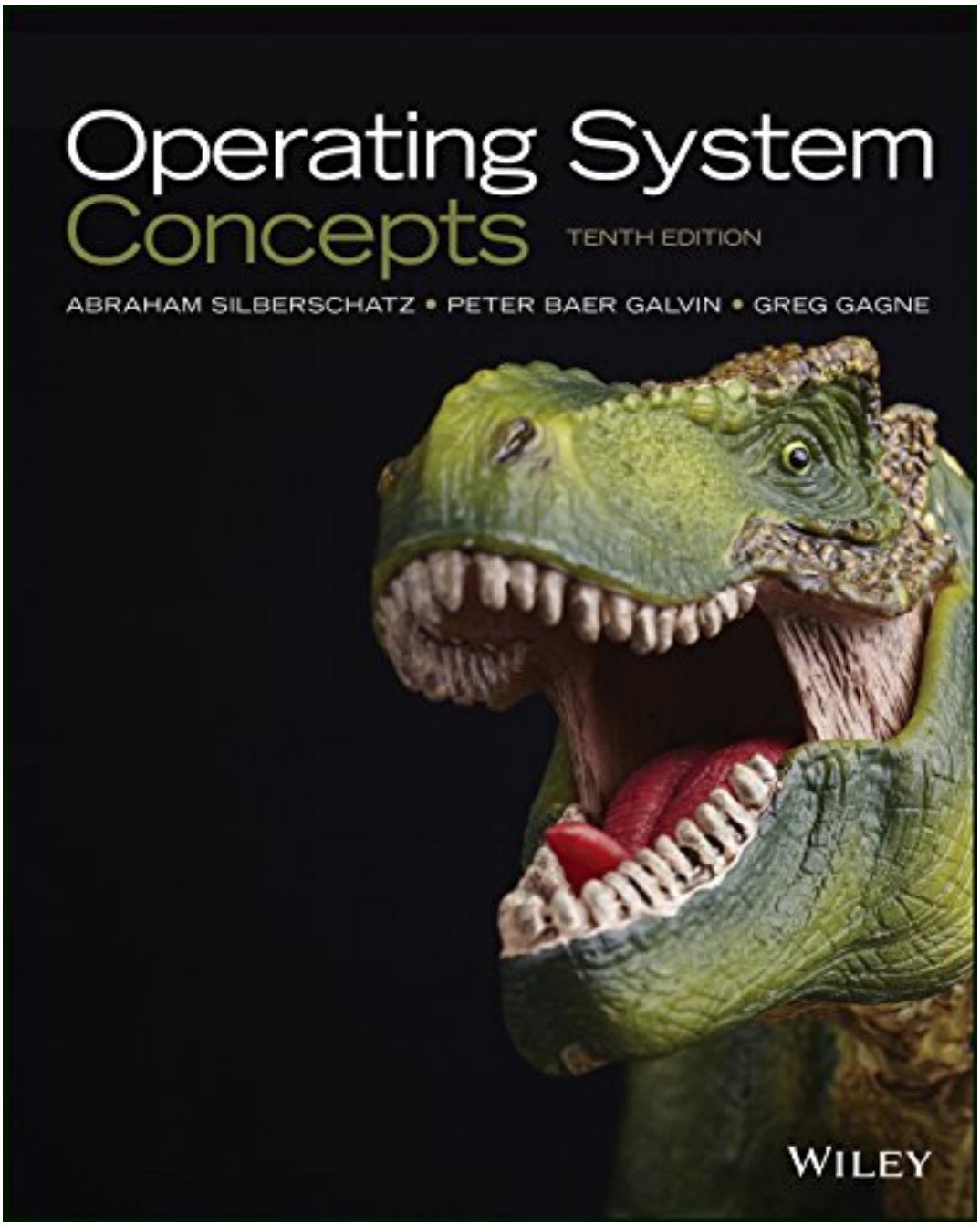
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Operating System Concepts

TENTH EDITION

ABRAHAM SILBERSCHATZ • PETER BAER GALVIN • GREG GAGNE



WILEY

Preface

Operating systems are an essential part of any computer system. Similarly, a course on operating systems is an essential part of any computer science education. This field is undergoing rapid change, as computers are now prevalent in virtually every arena of day-to-day life—from embedded devices in automobiles through the most sophisticated planning tools for governments and multinational firms. Yet the fundamental concepts remain fairly clear, and it is on these that we base this book.

We wrote this book as a text for an introductory course in operating systems at the junior or senior undergraduate level or at the first-year graduate level. We hope that practitioners will also find it useful. It provides a clear description of the *concepts* that underlie operating systems. As prerequisites, we assume that the reader is familiar with basic data structures, computer organization, and a high-level language, such as C or Java. The hardware topics required for an understanding of operating systems are covered in Chapter 1. In that chapter, we also include an overview of the fundamental data structures that are prevalent in most operating systems. For code examples, we use predominantly C, as well as a significant amount of Java, but the reader can still understand the algorithms without a thorough knowledge of these languages.

Concepts are presented using intuitive descriptions. Important theoretical results are covered, but formal proofs are largely omitted. The bibliographical notes at the end of each chapter contain pointers to research papers in which results were first presented and proved, as well as references to recent material for further reading. In place of proofs, figures and examples are used to suggest why we should expect the result in question to be true.

The fundamental concepts and algorithms covered in the book are often based on those used in both open-source and commercial operating systems. Our aim is to present these concepts and algorithms in a general setting that is not tied to one particular operating system. However, we present a large number of examples that pertain to the most popular and the most innovative operating systems, including Linux, Microsoft Windows, Apple macOS (the original name, OS X, was changed in 2016 to match the naming scheme of other Apple products), and Solaris. We also include examples of both Android and iOS, currently the two dominant mobile operating systems.

The organization of the text reflects our many years of teaching courses on operating systems. Consideration was also given to the feedback provided

by the reviewers of the text, along with the many comments and suggestions we received from readers of our previous editions and from our current and former students. This Tenth Edition also reflects most of the curriculum guidelines in the operating-systems area in *Computer Science Curricula 2013*, the most recent curriculum guidelines for undergraduate degree programs in computer science published by the IEEE Computing Society and the Association for Computing Machinery (ACM).

What's New in This Edition

For the Tenth Edition, we focused on revisions and enhancements aimed at lowering costs to the students, better engaging them in the learning process, and providing increased support for instructors.

According to the publishing industry's most trusted market research firm, Outsell, 2015 represented a turning point in text usage: for the first time, student preference for digital learning materials was higher than for print, and the increase in preference for digital has been accelerating since.

While print remains important for many students as a pedagogical tool, the Tenth Edition is being delivered in forms that emphasize support for learning from digital materials. All forms we are providing dramatically reduce the cost to students compared to the Ninth Edition. These forms are:

- **Stand-alone e-text now with significan enhancements**. The e-text format for the Tenth Edition adds exercises with solutions at the ends of main sections, hide/reveal definitions for key terms, and a number of animated figures. It also includes additional "Practice Exercises" with solutions for each chapter, extra exercises, programming problems and projects, "Further Reading" sections, a complete glossary, and four appendices for legacy operating systems.
- **E-text with print companion bundle**. For a nominal additional cost, the e-text also is available with an abridged print companion that includes a loose-leaf copy of the main chapter text, end-of-chapter "Practice Exercises" (solutions available online), and "Further Reading" sections. Instructors may also order bound print companions for the bundled package by contacting their Wiley account representative.

Although we highly encourage all instructors and students to take advantage of the cost, content, and learning advantages of the e-text edition, it is possible for instructors to work with their Wiley Account Manager to create a custom print edition.

To explore these options further or to discuss other options, contact your Wiley account manager (http://www.wiley.com/go/whosmyrep) or visit the product information page for this text on wiley.com

Book Material

The book consists of 21 chapters and 4 appendices. Each chapter and appendix contains the text, as well as the following enhancements:

- A set of practice exercises, including solutions
- A set of regular exercises
- A set of programming problems
- A set of programming projects
- A Further Reading section
- Pop-up definitions of important (blue) terms
- A glossary of important terms
- Animations that describe specific key concepts

A hard copy of the text is available in book stores and online. That version has the same text chapters as the electronic version. It does not, however, include the appendices, the regular exercises, the solutions to the practice exercises, the programming problems, the programming projects, and some of the other enhancements found in this ePub electronic book.

Content of This Book

The text is organized in ten major parts:

- **Overview**. Chapters 1 and 2 explain what operating systems are, what they do, and how they are designed and constructed. These chapters discuss what the common features of an operating system are and what an operating system does for the user. We include coverage of both traditional PC and server operating systems and operating systems for mobile devices. The presentation is motivational and explanatory in nature. We have avoided a discussion of how things are done internally in these chapters. Therefore, they are suitable for individual readers or for students in lower-level classes who want to learn what an operating system is without getting into the details of the internal algorithms.
- **Process management**. Chapters 3 through 5 describe the process concept and concurrency as the heart of modern operating systems. A ***process*** is the unit of work in a system. Such a system consists of a collection of ***concurrently*** executing processes, some executing operating-system code and others executing user code. These chapters cover methods for process scheduling and interprocess communication. Also included is a detailed discussion of threads, as well as an examination of issues related to multicore systems and parallel programming.
- **Process synchronization**. Chapters 6 through 8 cover methods for process synchronization and deadlock handling. Because we have increased the coverage of process synchronization, we have divided the former Chapter 5 (Process Synchronization) into two separate chapters: Chapter 6, Synchronization Tools, and Chapter 7, Synchronization Examples.
- **Memory management**. Chapters 9 and 10 deal with the management of main memory during the execution of a process. To improve both the

utilization of the CPU and the speed of its response to its users, the computer must keep several processes in memory. There are many different memory-management schemes, reflecting various approaches to memory management, and the effectiveness of a particular algorithm depends on the situation.

- **Storage management**. Chapters 11 and 12 describe how mass storage and I/O are handled in a modern computer system. The I/O devices that attach to a computer vary widely, and the operating system needs to provide a wide range of functionality to applications to allow them to control all aspects of these devices. We discuss system I/O in depth, including I/O system design, interfaces, and internal system structures and functions. In many ways, I/O devices are the slowest major components of the computer. Because they represent a performance bottleneck, we also examine performance issues associated with I/O devices.
- **File systems**. Chapters 13 through 15 discuss how file systems are handled in a modern computer system. File systems provide the mechanism for online storage of and access to both data and programs. We describe the classic internal algorithms and structures of storage management and provide a firm practical understanding of the algorithms used—their properties, advantages, and disadvantages.
- **Security and protection**. Chapters 16 and 17 discuss the mechanisms necessary for the security and protection of computer systems. The processes in an operating system must be protected from one another's activities. To provide such protection, we must ensure that only processes that have gained proper authorization from the operating system can operate on the files, memory, CPU, and other resources of the system. Protection is a mechanism for controlling the access of programs, processes, or users to computer-system resources. This mechanism must provide a means of specifying the controls to be imposed, as well as a means of enforcement. Security protects the integrity of the information stored in the system (both data and code), as well as the physical resources of the system, from unauthorized access, malicious destruction or alteration, and accidental introduction of inconsistency.
- **Advanced topics**. Chapters 18 and 19 discuss virtual machines and networks/distributed systems. Chapter 18 provides an overview of virtual machines and their relationship to contemporary operating systems. Included is a general description of the hardware and software techniques that make virtualization possible. Chapter 19 provides an overview of computer networks and distributed systems, with a focus on the Internet and TCP/IP.
- **Case studies**. Chapter 20 and 21 present detailed case studies of two real operating systems—Linux and Windows 10.
- **Appendices**. Appendix A discusses several old influential operating systems that are no longer in use. Appendices B through D cover in great detaisl three older operating systems— Windows 7, BSD, and Mach.

Programming Environments

The text provides several example programs written in C and Java. These programs are intended to run in the following programming environments:

- **POSIX**. POSIX (which stands for *Portable Operating System Interface*) represents a set of standards implemented primarily for UNIX-based operating systems. Although Windows systems can also run certain POSIX programs, our coverage of POSIX focuses on Linux and UNIX systems. POSIX-compliant systems must implement the POSIX core standard (POSIX.1); Linux and macOS are examples of POSIX-compliant systems. POSIX also defines several extensions to the standards, including real-time extensions (POSIX.1b) and an extension for a threads library (POSIX.1c, better known as Pthreads). We provide several programming examples written in C illustrating the POSIX base API, as well as Pthreads and the extensions for real-time programming. These example programs were tested on Linux 4.4 and macOS 10.11 systems using the `gcc` compiler.
- **Java**. Java is a widely used programming language with a rich API and built-in language support for concurrent and parallel programming. Java programs run on any operating system supporting a Java virtual machine (or JVM). We illustrate various operating-system and networking concepts with Java programs tested using Version 1.8 of the Java Development Kit (JDK).
- **Windows systems**. The primary programming environment for Windows systems is the Windows API, which provides a comprehensive set of functions for managing processes, threads, memory, and peripheral devices. We supply a modest number of C programs illustrating the use of this API. Programs were tested on a system running Windows 10.

We have chosen these three programming environments because we believe that they best represent the two most popular operating-system models—Linux/UNIX and Windows—along with the widely used Java environment. Most programming examples are written in C, and we expect readers to be comfortable with this language. Readers familiar with both the C and Java languages should easily understand most programs provided in this text.

In some instances—such as thread creation—we illustrate a specific concept using all three programming environments, allowing the reader to contrast the three different libraries as they address the same task. In other situations, we may use just one of the APIs to demonstrate a concept. For example, we illustrate shared memory using just the POSIX API; socket programming in TCP/IP is highlighted using the Java API.

Linux Virtual Machine

To help students gain a better understanding of the Linux system, we provide a Linux virtual machine running the Ubuntu distribution with this text. The virtual machine, which is available for download from the text website

(http://www.os-book.com), also provides development environments including the gcc and Java compilers. Most of the programming assignments in the book can be completed using this virtual machine, with the exception of assignments that require the Windows API. The virtual machine can be installed and run on any host operating system that can run the VirtualBox virtualization software, which currently includes Windows 10 Linux, and macOS.

The Tenth Edition

As we wrote this Tenth Edition of *Operating System Concepts*, we were guided by the sustained growth in four fundamental areas that affect operating systems:

1. Mobile operating systems
2. Multicore systems
3. Virtualization
4. Nonvolatile memory secondary storage

To emphasize these topics, we have integrated relevant coverage throughout this new edition. For example, we have greatly increased our coverage of the Android and iOS mobile operating systems, as well as our coverage of the ARMv8 architecture that dominates mobile devices. We have also increased our coverage of multicore systems, including increased coverage of APIs that provide support for concurrency and parallelism. Nonvolatile memory devices like SSDs are now treated as the equals of hard-disk drives in the chapters that discuss I/O, mass storage, and file systems.

Several of our readers have expressed support for an increase in Java coverage, and we have provided additional Java examples throughout this edition.

Additionally, we have rewritten material in almost every chapter by bringing older material up to date and removing material that is no longer interesting or relevant. We have reordered many chapters and have, in some instances, moved sections from one chapter to another. We have also greatly revised the artwork, creating several new figures as well as modifying many existing figures.

Major Changes

The Tenth Edition update encompasses much more material than previous updates, in terms of both content and new supporting material. Next, we provide a brief outline of the major content changes in each chapter:

- **Chapter 1: Introduction** includes updated coverage of multicore systems, as well as new coverage of NUMA systems and Hadoop clusters. Old material has been updated, and new motivation has been added for the study of operating systems.
- **Chapter 2: Operating-System Structures** provides a significantly revised discussion of the design and implementation of operating systems. We have updated our treatment of Android and iOS and have revised our

coverage of the system boot process with a focus on GRUB for Linux systems. New coverage of the Windows subsystem for Linux is included as well. We have added new sections on linkers and loaders, and we now discuss why applications are often operating-system specific. Finally, we have added a discussion of the BCC debugging toolset.

- **Chapter 3: Processes** simplifies the discussion of scheduling so that it now includes only CPU scheduling issues. New coverage describes the memory layout of a C program, the Android process hierarchy, Mach message passing, and Android RPCs. We have also replaced coverage of the traditional UNIX/Linux `init` process with coverage of `systemd`.
- **Chapter 4: Threads and Concurrency** (previously Threads) increases the coverage of support for concurrent and parallel programming at the API and library level. We have revised the section on Java threads so that it now includes futures and have updated the coverage of Apple's Grand Central Dispatch so that it now includes Swift. New sections discuss fork-join parallelism using the fork-join framework in Java, as well as Intel thread building blocks.
- **Chapter 5: CPU Scheduling** (previously Chapter 6) revises the coverage of multilevel queue and multicore processing scheduling. We have integrated coverage of NUMA-aware scheduling issues throughout, including how this scheduling affects load balancing. We also discuss related modifications to the Linux CFS scheduler. New coverage combines discussions of round-robin and priority scheduling, heterogeneous multiprocessing, and Windows 10 scheduling.
- **Chapter 6: Synchronization Tools** (previously part of Chapter 5, Process Synchronization) focuses on various tools for synchronizing processes. Significant new coverage discusses architectural issues such as instruction reordering and delayed writes to buffers. The chapter also introduces lock-free algorithms using compare-and-swap (CAS) instructions. No specific APIs are presented; rather, the chapter provides an introduction to race conditions and general tools that can be used to prevent data races. Details include new coverage of memory models, memory barriers, and liveness issues.
- **Chapter 7: Synchronization Examples** (previously part of Chapter 5, Process Synchronization) introduces classical synchronization problems and discusses specific API support for designing solutions that solve these problems. The chapter includes new coverage of POSIX named and unnamed semaphores, as well as condition variables. A new section on Java synchronization is included as well.
- **Chapter 8: Deadlocks** (previously Chapter 7) provides minor updates, including a new section on livelock and a discussion of deadlock as an example of a liveness hazard. The chapter includes new coverage of the Linux `lockdep` and the BCC `deadlock_detector` tools, as well as coverage of Java deadlock detection using thread dumps.
- **Chapter 9: Main Memory** (previously Chapter 8) includes several revisions that bring the chapter up to date with respect to memory manage-

ment on modern computer systems. We have added new coverage of the ARMv8 64-bit architecture, updated the coverage of dynamic link libraries, and changed swapping coverage so that it now focuses on swapping pages rather than processes. We have also eliminated coverage of segmentation.

- **Chapter 10: Virtual Memory** (previously Chapter 9) contains several revisions, including updated coverage of memory allocation on NUMA systems and global allocation using a free-frame list. New coverage includes compressed memory, major/minor page faults, and memory management in Linux and Windows 10.
- **Chapter 11: Mass-Storage Structure** (previously Chapter 10) adds coverage of nonvolatile memory devices, such as flash and solid-state disks. Hard-drive scheduling is simplified to show only currently used algorithms. Also included are a new section on cloud storage, updated RAID coverage, and a new discussion of object storage.
- **Chapter 12, I/O** (previously Chapter 13) updates the coverage of technologies and performance numbers, expands the coverage of synchronous/asynchronous and blocking/nonblocking I/O, and adds a section on vectored I/O. It also expands coverage of power management for mobile operating systems.
- **Chapter 13: File-System Interface** (previously Chapter 11) has been updated with information about current technologies. In particular, the coverage of directory structures has been improved, and the coverage of protection has been updated. The memory-mapped files section has been expanded, and a Windows API example has been added to the discussion of shared memory. The ordering of topics is refactored in Chapter 13 and 14.
- **Chapter 14: File-System Implementation** (previously Chapter 12) has been updated with coverage of current technologies. The chapter now includes discussions of TRIM and the Apple File System. In addition, the discussion of performance has been updated, and the coverage of journaling has been expanded.
- **Chapter 15: File System Internals** is new and contains updated information from previous Chapters 11 and 12.
- **Chapter 16: Security** (previously Chapter 15) now precedes the protection chapter. It includes revised and updated terms for current security threats and solutions, including ransomware and remote access tools. The principle of least privilege is emphasized. Coverage of code-injection vulnerabilities and attacks has been revised and now includes code samples. Discussion of encryption technologies has been updated to focus on the technologies currently used. Coverage of authentication (by passwords and other methods) has been updated and expanded with helpful hints. Additions include a discussion of address-space layout randomization and a new summary of security defenses. The Windows 7 example has been updated to Windows 10.
- **Chapter 17: Protection** (previously Chapter 14) contains major changes. The discussion of protection rings and layers has been updated and now

refers to the Bell–LaPadula model and explores the ARM model of TrustZones and Secure Monitor Calls. Coverage of the need-to-know principle has been expanded, as has coverage of mandatory access control. Subsections on Linux capabilities, Darwin entitlements, security integrity protection, system-call filtering, sandboxing, and code signing have been added. Coverage of run-time-based enforcement in Java has also been added, including the stack inspection technique.

- **Chapter 18: Virtual Machines** (previously Chapter 16) includes added details about hardware assistance technologies. Also expanded is the topic of application containment, now including containers, zones, docker, and Kubernetes. A new section discusses ongoing virtualization research, including unikernels, library operating systems, partitioning hypervisors, and separation hypervisors.
- **Chapter 19, Networks and Distributed Systems** (previously Chapter 17) has been substantially updated and now combines coverage of computer networks and distributed systems. The material has been revised to bring it up to date with respect to contemporary computer networks and distributed systems. The TCP/IP model receives added emphasis, and a discussion of cloud storage has been added. The section on network topologies has been removed. Coverage of name resolution has been expanded and a Java example added. The chapter also includes new coverage of distributed file systems, including MapReduce on top of Google file system, Hadoop, GPFS, and Lustre.
- **Chapter 20: The Linux System** (previously Chapter 18) has been updated to cover the Linux 4.*i* kernel.
- **Chapter 21: The Windows 10 System** is a new chapter that covers the internals of Windows 10.
- **Appendix A: Influentia Operating Systems** has been updated to include material from chapters that are no longer covered in the text.

Supporting Website

When you visit the website supporting this text at http://www.os-book.com, you can download the following resources:

- Linux virtual machine
- C and Java source code
- The complete set of figures and illustrations
- FreeBSD, Mach, and Windows 7 case studies
- Errata
- Bibliography

Notes to Instructors

On the website for this text, we provide several sample syllabi that suggest various approaches for using the text in both introductory and advanced courses.

As a general rule, we encourage instructors to progress sequentially through the chapters, as this strategy provides the most thorough study of operating systems. However, by using the sample syllabi, an instructor can select a different ordering of chapters (or subsections of chapters).

In this edition, we have added many new written exercises and programming problems and projects. Most of the new programming assignments involve processes, threads, process scheduling, process synchronization, and memory management. Some involve adding kernel modules to the Linux system, which requires using either the Linux virtual machine that accompanies this text or another suitable Linux distribution.

Solutions to written exercises and programming assignments are available to instructors who have adopted this text for their operating-system class. To obtain these restricted supplements, contact your local John Wiley & Sons sales representative. You can find your Wiley representative by going to http://www.wiley.com/college and clicking "Who's my rep?"

Notes to Students

We encourage you to take advantage of the practice exercises that appear at the end of each chapter. We also encourage you to read through the study guide, which was prepared by one of our students. Finally, for students who are unfamiliar with UNIX and Linux systems, we recommend that you download and install the Linux virtual machine that we include on the supporting website. Not only will this provide you with a new computing experience, but the open-source nature of Linux will allow you to easily examine the inner details of this popular operating system. We wish you the very best of luck in your study of operating systems!

Contacting Us

We have endeavored to eliminate typos, bugs, and the like from the text. But, as in new releases of software, bugs almost surely remain. An up-to-date errata list is accessible from the book's website. We would be grateful if you would notify us of any errors or omissions in the book that are not on the current list of errata.

We would be glad to receive suggestions on improvements to the book. We also welcome any contributions to the book website that could be of use to other readers, such as programming exercises, project suggestions, on-line labs and tutorials, and teaching tips. E-mail should be addressed to os-book-authors@cs.yale.edu.

Acknowledgments

Many people have helped us with this Tenth Edition, as well as with the previous nine editions from which it is derived.

Tenth Edition

- Rick Farrow provided expert advice as a technical editor.
- Jonathan Levin helped out with coverage of mobile systems, protection, and security.
- Alex Ionescu updated the previous Windows 7 chapter to provide Chapter 21: Windows 10.
- Sarah Diesburg revised Chapter 19: Networks and Distributed Systems.
- Brendan Gregg provided guidance on the BCC toolset.
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- Robert Love provided updates to Chapter 20: The Linux System.
- Michael Shapiro helped with storage and I/O technology details.
- Richard West provided insight on areas of virtualization research.
- Clay Breshears helped with coverage of Intel thread-building blocks.
- Gerry Howser gave feedback on motivating the study of operating systems and also tried out new material in his class.
- Judi Paige helped with generating figures and presentation of slides.
- Jay Gagne and Audra Rissmeyer prepared new artwork for this edition.
- Owen Galvin provided technical editing for Chapter 11 and Chapter 12.
- Mark Wogahn has made sure that the software to produce this book (LaTeX and fonts) works properly.
- Ranjan Kumar Meher rewrote some of the LaTeX software used in the production of this new text.

Previous Editions

- **First three editions**. This book is derived from the previous editions, the first three of which were coauthored by James Peterson.
- **General contributions**. Others who helped us with previous editions include Hamid Arabnia, Rida Bazzi, Randy Bentson, David Black, Joseph Boykin, Jeff Brumfield, Gael Buckley, Roy Campbell, P. C. Capon, John Carpenter, Gil Carrick, Thomas Casavant, Bart Childs, Ajoy Kumar Datta, Joe Deck, Sudarshan K. Dhall, Thomas Doeppner, Caleb Drake, M. Rasit Eskicioğlu, Hans Flack, Robert Fowler, G. Scott Graham, Richard Guy, Max Hailperin, Rebecca Hartman, Wayne Hathaway, Christopher Haynes, Don Heller, Bruce Hillyer, Mark Holliday, Dean Hougen, Michael Huang, Ahmed Kamel, Morty Kewstel, Richard Kieburtz, Carol Kroll, Morty Kwestel, Thomas LeBlanc, John Leggett, Jerrold Leichter, Ted Leung, Gary Lippman, Carolyn Miller, Michael Molloy, Euripides Montagne, Yoichi Muraoka, Jim M. Ng, Banu Özden, Ed Posnak, Boris Putanec, Charles

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- **Specifi Contributions**
  - Robert Love updated both Chapter 20 and the Linux coverage throughout the text, as well as answering many of our Android-related questions.
  - Appendix B was written by Dave Probert and was derived from Chapter 22 of the Eighth Edition of *Operating System Concepts.*
  - Jonathan Katz contributed to Chapter 16. Richard West provided input into Chapter 18. Salahuddin Khan updated Section 16.7 to provide new coverage of Windows 7 security.
  - Parts of Chapter 19 were derived from a paper by Levy and Silberschatz [1990].
  - Chapter 20 was derived from an unpublished manuscript by Stephen Tweedie.
  - Cliff Martin helped with updating the UNIX appendix to cover FreeBSD.
  - Some of the exercises and accompanying solutions were supplied by Arvind Krishnamurthy.
  - Andrew DeNicola prepared the student study guide that is available on our website. Some of the slides were prepared by Marilyn Turnamian.
  - Mike Shapiro, Bryan Cantrill, and Jim Mauro answered several Solaris-related questions, and Bryan Cantrill from Sun Microsystems helped with the ZFS coverage. Josh Dees and Rob Reynolds contributed coverage of Microsoft's NET.
  - Owen Galvin helped copy-edit Chapter 18 edition.

Book Production

The Executive Editor was Don Fowley. The Senior Production Editor was Ken Santor. The Freelance Developmental Editor was Chris Nelson. The Assistant Developmental Editor was Ryann Dannelly. The cover designer was Tom Nery. The copyeditor was Beverly Peavler. The freelance proofreader was Katrina Avery. The freelance indexer was WordCo, Inc. The Aptara LaTex team consisted of Neeraj Saxena and Lav kush.

Personal Notes

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Peter would like to thank his wife Carla and his children, Gwen, Owen, and Maddie.

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Abraham Silberschatz, New Haven, CT
Peter Baer Galvin, Boston, MA
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Part One

Overview

An *operating system* acts as an intermediary between the user of a computer and the computer hardware. The purpose of an operating system is to provide an environment in which a user can execute programs in a *convenient* and *efficient* manner.

An operating system is software that manages the computer hardware. The hardware must provide appropriate mechanisms to ensure the correct operation of the computer system and to prevent programs from interfering with the proper operation of the system.

Internally, operating systems vary greatly in their makeup, since they are organized along many different lines. The design of a new operating system is a major task, and it is important that the goals of the system be well defined before the design begins.

Because an operating system is large and complex, it must be created piece by piece. Each of these pieces should be a well-delineated portion of the system, with carefully defined inputs, outputs, and functions.

CHAPTER 1

Introduction



An **operating system** is software that manages a computer's hardware. It also provides a basis for application programs and acts as an intermediary between the computer user and the computer hardware. An amazing aspect of operating systems is how they vary in accomplishing these tasks in a wide variety of computing environments. Operating systems are everywhere, from cars and home appliances that include "Internet of Things" devices, to smart phones, personal computers, enterprise computers, and cloud computing environments.

In order to explore the role of an operating system in a modern computing environment, it is important first to understand the organization and architecture of computer hardware. This includes the CPU, memory, and I/O devices, as well as storage. A fundamental responsibility of an operating system is to allocate these resources to programs.

Because an operating system is large and complex, it must be created piece by piece. Each of these pieces should be a well-delineated portion of the system, with carefully defined inputs, outputs, and functions. In this chapter, we provide a general overview of the major components of a contemporary computer system as well as the functions provided by the operating system. Additionally, we cover several topics to help set the stage for the remainder of the text: data structures used in operating systems, computing environments, and open-source and free operating systems.

CHAPTER OBJECTIVES

- Describe the general organization of a computer system and the role of interrupts.
- Describe the components in a modern multiprocessor computer system.
- Illustrate the transition from user mode to kernel mode.
- Discuss how operating systems are used in various computing environments.
- Provide examples of free and open-source operating systems.

1.1 What Operating Systems Do

We begin our discussion by looking at the operating system's role in the overall computer system. A computer system can be divided roughly into four components: the ***hardware***, the ***operating system***, the ***application programs***, and a ***user*** (Figure 1.1).

The **hardware**—the central processing unit (CPU), the memory, and the input/output (I/O) devices—provides the basic computing resources for the system. The **application programs**—such as word processors, spreadsheets, compilers, and web browsers—define the ways in which these resources are used to solve users' computing problems. The operating system controls the hardware and coordinates its use among the various application programs for the various users.

We can also view a computer system as consisting of hardware, software, and data. The operating system provides the means for proper use of these resources in the operation of the computer system. An operating system is similar to a government. Like a government, it performs no useful function by itself. It simply provides an ***environment*** within which other programs can do useful work.

To understand more fully the operating system's role, we next explore operating systems from two viewpoints: that of the user and that of the system.

1.1.1 User View

The user's view of the computer varies according to the interface being used. Many computer users sit with a laptop or in front of a PC consisting of a monitor, keyboard, and mouse. Such a system is designed for one user to monopolize its resources. The goal is to maximize the work (or play) that the user is performing. In this case, the operating system is designed mostly for **ease of use**, with some attention paid to performance and security and none paid to **resource utilization**—how various hardware and software resources are shared.

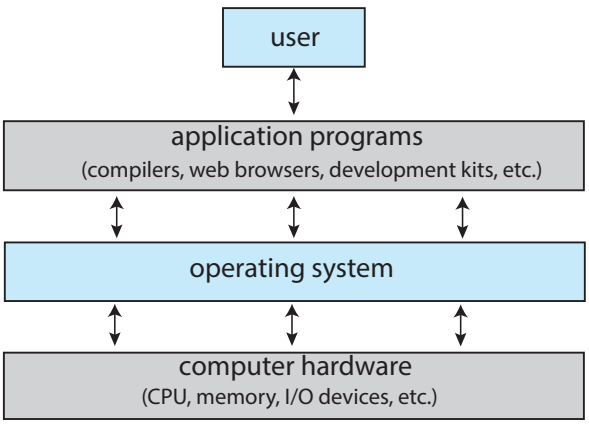


Figure 1.1 Abstract view of the components of a computer system.

Increasingly, many users interact with mobile devices such as smartphones and tablets—devices that are replacing desktop and laptop computer systems for some users. These devices are typically connected to networks through cellular or other wireless technologies. The user interface for mobile computers generally features a **touch screen**, where the user interacts with the system by pressing and swiping fingers across the screen rather than using a physical keyboard and mouse. Many mobile devices also allow users to interact through a **voice recognition** interface, such as Apple's **Siri**.

Some computers have little or no user view. For example, **embedded computers** in home devices and automobiles may have numeric keypads and may turn indicator lights on or off to show status, but they and their operating systems and applications are designed primarily to run without user intervention.

1.1.2 System View

From the computer's point of view, the operating system is the program most intimately involved with the hardware. In this context, we can view an operating system as a **resource allocator**. A computer system has many resources that may be required to solve a problem: CPU time, memory space, storage space, I/O devices, and so on. The operating system acts as the manager of these resources. Facing numerous and possibly conflicting requests for resources, the operating system must decide how to allocate them to specific programs and users so that it can operate the computer system efficiently and fairly.

A slightly different view of an operating system emphasizes the need to control the various I/O devices and user programs. An operating system is a control program. A **control program** manages the execution of user programs to prevent errors and improper use of the computer. It is especially concerned with the operation and control of I/O devices.

1.1.3 Defining Operating Systems

By now, you can probably see that the term ***operating system*** covers many roles and functions. That is the case, at least in part, because of the myriad designs and uses of computers. Computers are present within toasters, cars, ships, spacecraft, homes, and businesses. They are the basis for game machines, cable TV tuners, and industrial control systems.

To explain this diversity, we can turn to the history of computers. Although computers have a relatively short history, they have evolved rapidly. Computing started as an experiment to determine what could be done and quickly moved to fixed-purpose systems for military uses, such as code breaking and trajectory plotting, and governmental uses, such as census calculation. Those early computers evolved into general-purpose, multifunction mainframes, and that's when operating systems were born. In the 1960s, **Moore's Law** predicted that the number of transistors on an integrated circuit would double every 18 months, and that prediction has held true. Computers gained in functionality and shrank in size, leading to a vast number of uses and a vast number and variety of operating systems. (See Appendix A for more details on the history of operating systems.)

How, then, can we define what an operating system is? In general, we have no completely adequate definition of an operating system. Operating systems

exist because they offer a reasonable way to solve the problem of creating a usable computing system. The fundamental goal of computer systems is to execute programs and to make solving user problems easier. Computer hardware is constructed toward this goal. Since bare hardware alone is not particularly easy to use, application programs are developed. These programs require certain common operations, such as those controlling the I/O devices. The common functions of controlling and allocating resources are then brought together into one piece of software: the operating system.

In addition, we have no universally accepted definition of what is part of the operating system. A simple viewpoint is that it includes everything a vendor ships when you order "the operating system." The features included, however, vary greatly across systems. Some systems take up less than a megabyte of space and lack even a full-screen editor, whereas others require gigabytes of space and are based entirely on graphical windowing systems. A more common definition, and the one that we usually follow, is that the operating system is the one program running at all times on the computer—usually called the **kernel**. Along with the kernel, there are two other types of programs: **system programs**, which are associated with the operating system but are not necessarily part of the kernel, and application programs, which include all programs not associated with the operation of the system.

The matter of what constitutes an operating system became increasingly important as personal computers became more widespread and operating systems grew increasingly sophisticated. In 1998, the United States Department of Justice filed suit against Microsoft, in essence claiming that Microsoft included too much functionality in its operating systems and thus prevented application vendors from competing. (For example, a web browser was an integral part of Microsoft's operating systems.) As a result, Microsoft was found guilty of using its operating-system monopoly to limit competition.

Today, however, if we look at operating systems for mobile devices, we see that once again the number of features constituting the operating system is increasing. Mobile operating systems often include not only a core kernel but also **middleware**—a set of software frameworks that provide additional services to application developers. For example, each of the two most prominent mobile operating systems—Apple's iOS and Google's Android—features

WHY STUDY OPERATING SYSTEMS?

Although there are many practitioners of computer science, only a small percentage of them will be involved in the creation or modification of an operating system. Why, then, study operating systems and how they work? Simply because, as almost all code runs on top of an operating system, knowledge of how operating systems work is crucial to proper, efficient, effective, and secure programming. Understanding the fundamentals of operating systems, how they drive computer hardware, and what they provide to applications is not only essential to those who program them but also highly useful to those who write programs on them and use them.

a core kernel along with middleware that supports databases, multimedia, and graphics (to name only a few).

In summary, for our purposes, the operating system includes the always-running kernel, middleware frameworks that ease application development and provide features, and system programs that aid in managing the system while it is running. Most of this text is concerned with the kernel of general-purpose operating systems, but other components are discussed as needed to fully explain operating system design and operation.

1.2 Computer-System Organization

A modern general-purpose computer system consists of one or more CPUs and a number of device controllers connected through a common **bus** that provides access between components and shared memory (Figure 1.2). Each device controller is in charge of a specific type of device (for example, a disk drive, audio device, or graphics display). Depending on the controller, more than one device may be attached. For instance, one system USB port can connect to a USB hub, to which several devices can connect. A device controller maintains some local buffer storage and a set of special-purpose registers. The device controller is responsible for moving the data between the peripheral devices that it controls and its local buffer storage.

Typically, operating systems have a **device driver** for each device controller. This device driver understands the device controller and provides the rest of the operating system with a uniform interface to the device. The CPU and the device controllers can execute in parallel, competing for memory cycles. To ensure orderly access to the shared memory, a memory controller synchronizes access to the memory.

In the following subsections, we describe some basics of how such a system operates, focusing on three key aspects of the system. We start with interrupts, which alert the CPU to events that require attention. We then discuss storage structure and I/O structure.

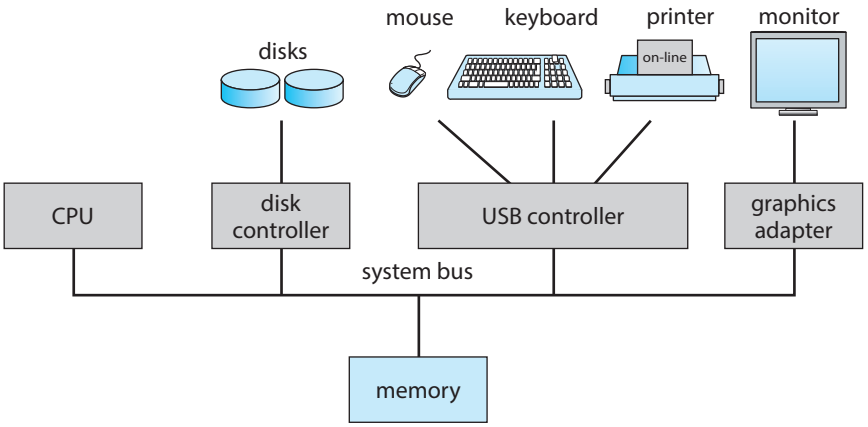
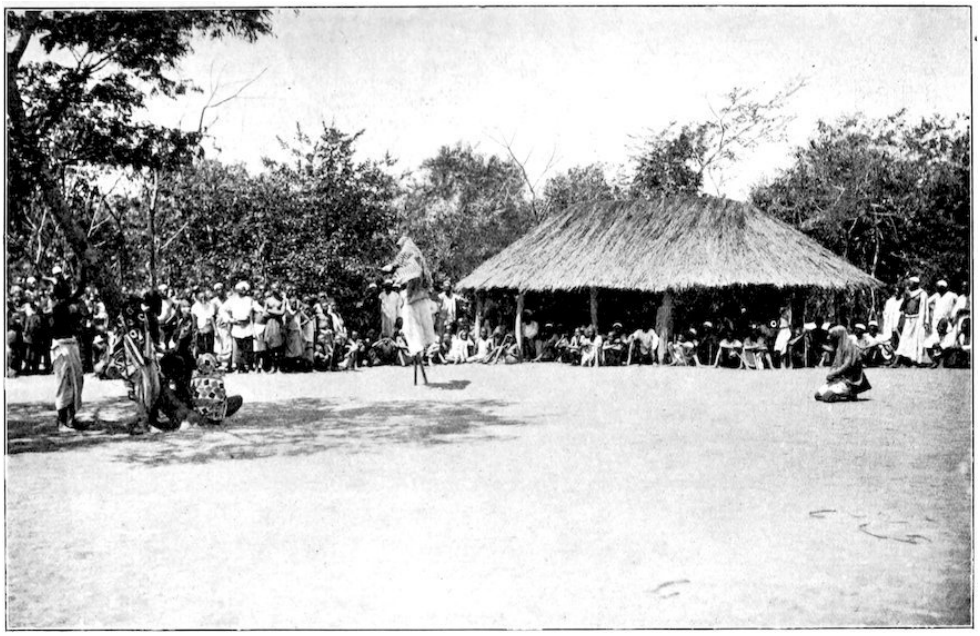


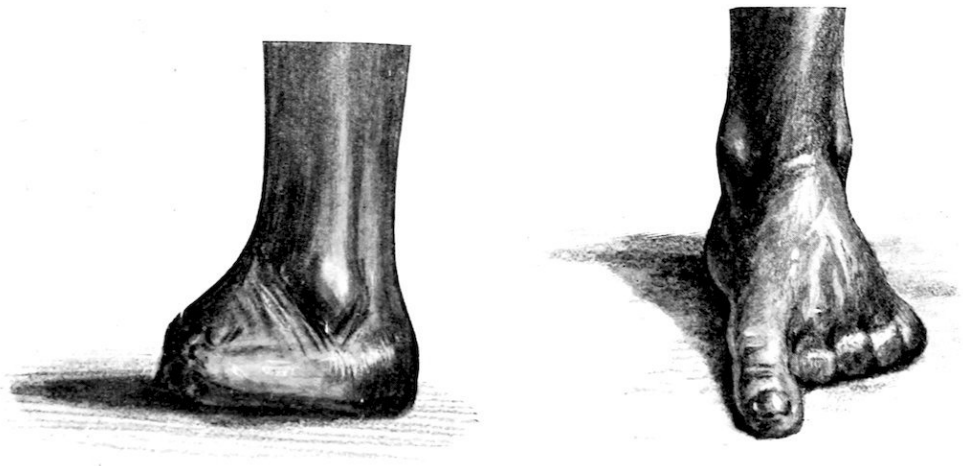
Figure 1.2 A typical PC computer system.

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DANCE ON STILTS AT THE GIRLS' UNYAGO, NIUCHI

Newala, too, suffers from the distance of its water-supply—at least the Newala of to-day does; there was once another Newala in a lovely valley at the foot of the plateau. I visited it and found scarcely a trace of houses, only a Christian cemetery, with the graves of several missionaries and their converts, remaining as a monument of its former glories. But the surroundings are wonderfully beautiful. A thick grove of splendid mango-trees closes in the weather-worn crosses and headstones; behind them, combining the useful and the agreeable, is a whole plantation of lemon-trees covered with ripe fruit; not the small African kind, but a much larger and also juicier imported variety, which drops into the hands of the passing traveller, without calling for any exertion on his part. Old Newala is now under the jurisdiction of the native pastor, Daudi, at Chingulungulu, who, as I am on very friendly terms with him, allows me, as a matter of course, the use of this lemon-grove during my stay at Newala.



FEET MUTILATED BY THE RAVAGES OF THE "JIGGER"
(*Sarcopsylla penetrans*)

The water-supply of New Newala is in the bottom of the valley, some 1,600 feet lower down. The way is not only long and fatiguing, but the water, when we get it, is thoroughly bad. We are suffering not only from this, but from the fact that the arrangements at Newala are nothing short of luxurious. We have a separate kitchen—a hut built against the *boma* palisade on the right of the *baraza*, the interior of which is not visible from our usual position. Our two cooks were not long in finding this out, and they consequently do—or rather neglect to do—what they please. In any case they do not seem to be very particular about the boiling of our drinking-water—at least I can attribute to no other cause certain attacks of a dysenteric nature, from which both Knudsen and I have suffered for some time. If a man like Omari has to be left unwatched for a moment, he is capable of anything. Besides this complaint, we are inconvenienced by the state of our nails, which have become as hard as glass, and crack on the slightest provocation, and I have the additional infliction of pimples all over me. As if all this were not enough, we have also, for the last week been waging war against the jigger, who has found his Eldorado in the hot sand of the Makonde plateau. Our men are seen all day long—whenever their chronic colds and the dysentery likewise raging among them permit—occupied in removing this scourge of Africa from their feet and trying to prevent the disastrous consequences of its presence. It is quite common to see natives of this place with one or two toes missing; many have lost all their toes,

or even the whole front part of the foot, so that a well-formed leg ends in a shapeless stump. These ravages are caused by the female of *Sarcopsylla penetrans*, which bores its way under the skin and there develops an egg-sac the size of a pea. In all books on the subject, it is stated that one's attention is called to the presence of this parasite by an intolerable itching. This agrees very well with my experience, so far as the softer parts of the sole, the spaces between and under the toes, and the side of the foot are concerned, but if the creature penetrates through the harder parts of the heel or ball of the foot, it may escape even the most careful search till it has reached maturity. Then there is no time to be lost, if the horrible ulceration, of which we see cases by the dozen every day, is to be prevented. It is much easier, by the way, to discover the insect on the white skin of a European than on that of a native, on which the dark speck scarcely shows. The four or five jiggers which, in spite of the fact that I constantly wore high laced boots, chose my feet to settle in, were taken out for me by the all-accomplished Knudsen, after which I thought it advisable to wash out the cavities with corrosive sublimate. The natives have a different sort of disinfectant—they fill the hole with scraped roots. In a tiny Makua village on the slope of the plateau south of Newala, we saw an old woman who had filled all the spaces under her toe-nails with powdered roots by way of prophylactic treatment. What will be the result, if any, who can say?

The rest of the many trifling ills which trouble our existence are really more comic than serious. In the absence of anything else to smoke, Knudsen and I at last opened a box of cigars procured from the Indian store-keeper at Lindi, and tried them, with the most distressing results. Whether they contain opium or some other narcotic, neither of us can say, but after the tenth puff we were both "off," three-quarters stupefied and unspeakably wretched. Slowly we recovered—and what happened next? Half-an-hour later we were once more smoking these poisonous concoctions—so insatiable is the craving for tobacco in the tropics.

Even my present attacks of fever scarcely deserve to be taken seriously. I have had no less than three here at Newala, all of which have run their course in an incredibly short time. In the early afternoon, I am busy with my old natives, asking questions and making notes. The strong midday coffee has stimulated my spirits to

an extraordinary degree, the brain is active and vigorous, and work progresses rapidly, while a pleasant warmth pervades the whole body. Suddenly this gives place to a violent chill, forcing me to put on my overcoat, though it is only half-past three and the afternoon sun is at its hottest. Now the brain no longer works with such acuteness and logical precision; more especially does it fail me in trying to establish the syntax of the difficult Makua language on which I have ventured, as if I had not enough to do without it. Under the circumstances it seems advisable to take my temperature, and I do so, to save trouble, without leaving my seat, and while going on with my work. On examination, I find it to be 101·48°. My tutors are abruptly dismissed and my bed set up in the *baraza*; a few minutes later I am in it and treating myself internally with hot water and lemon-juice.

Three hours later, the thermometer marks nearly 104°, and I make them carry me back into the tent, bed and all, as I am now perspiring heavily, and exposure to the cold wind just beginning to blow might mean a fatal chill. I lie still for a little while, and then find, to my great relief, that the temperature is not rising, but rather falling. This is about 7.30 p.m. At 8 p.m. I find, to my unbounded astonishment, that it has fallen below 98·6°, and I feel perfectly well. I read for an hour or two, and could very well enjoy a smoke, if I had the wherewithal—Indian cigars being out of the question.

Having no medical training, I am at a loss to account for this state of things. It is impossible that these transitory attacks of high fever should be malarial; it seems more probable that they are due to a kind of sunstroke. On consulting my note-book, I become more and more inclined to think this is the case, for these attacks regularly follow extreme fatigue and long exposure to strong sunshine. They at least have the advantage of being only short interruptions to my work, as on the following morning I am always quite fresh and fit. My treasure of a cook is suffering from an enormous hydrocele which makes it difficult for him to get up, and Moritz is obliged to keep in the dark on account of his inflamed eyes. Knudsen's cook, a raw boy from somewhere in the bush, knows still less of cooking than Omari; consequently Nils Knudsen himself has been promoted to the vacant post. Finding that we had come to the end of our supplies, he began by sending to Chingulungulu for the four sucking-pigs which we had

bought from Matola and temporarily left in his charge; and when they came up, neatly packed in a large crate, he callously slaughtered the biggest of them. The first joint we were thoughtless enough to entrust for roasting to Knudsen's *mshenzi* cook, and it was consequently uneatable; but we made the rest of the animal into a jelly which we ate with great relish after weeks of underfeeding, consuming incredible helpings of it at both midday and evening meals. The only drawback is a certain want of variety in the tinned vegetables. Dr. Jäger, to whom the Geographical Commission entrusted the provisioning of the expeditions—mine as well as his own—because he had more time on his hands than the rest of us, seems to have laid in a huge stock of Teltow turnips,[46] an article of food which is all very well for occasional use, but which quickly palls when set before one every day; and we seem to have no other tins left. There is no help for it—we must put up with the turnips; but I am certain that, once I am home again, I shall not touch them for ten years to come.

Amid all these minor evils, which, after all, go to make up the genuine flavour of Africa, there is at least one cheering touch: Knudsen has, with the dexterity of a skilled mechanic, repaired my 9 × 12 cm. camera, at least so far that I can use it with a little care. How, in the absence of finger-nails, he was able to accomplish such a ticklish piece of work, having no tool but a clumsy screw-driver for taking to pieces and putting together again the complicated mechanism of the instantaneous shutter, is still a mystery to me; but he did it successfully. The loss of his finger-nails shows him in a light contrasting curiously enough with the intelligence evinced by the above operation; though, after all, it is scarcely surprising after his ten years' residence in the bush. One day, at Lindi, he had occasion to wash a dog, which must have been in need of very thorough cleansing, for the bottle handed to our friend for the purpose had an extremely strong smell. Having performed his task in the most conscientious manner, he perceived with some surprise that the dog did not appear much the better for it, and was further surprised by finding his own nails ulcerating away in the course of the next few days. "How was I to know that carbolic acid has to be diluted?" he mutters indignantly, from time to time, with a troubled gaze at his mutilated finger-tips.

Since we came to Newala we have been making excursions in all directions through the surrounding country, in accordance with old habit, and also because the *akida* Sefu did not get together the tribal elders from whom I wanted information so speedily as he had promised. There is, however, no harm done, as, even if seen only from the outside, the country and people are interesting enough.

The Makonde plateau is like a large rectangular table rounded off at the corners. Measured from the Indian Ocean to Newala, it is about seventy-five miles long, and between the Rovuma and the Lukuledi it averages fifty miles in breadth, so that its superficial area is about two-thirds of that of the kingdom of Saxony. The surface, however, is not level, but uniformly inclined from its south-western edge to the ocean. From the upper edge, on which Newala lies, the eye ranges for many miles east and north-east, without encountering any obstacle, over the Makonde bush. It is a green sea, from which here and there thick clouds of smoke rise, to show that it, too, is inhabited by men who carry on their tillage like so many other primitive peoples, by cutting down and burning the bush, and manuring with the ashes. Even in the radiant light of a tropical day such a fire is a grand sight.

Much less effective is the impression produced just now by the great western plain as seen from the edge of the plateau. As often as time permits, I stroll along this edge, sometimes in one direction, sometimes in another, in the hope of finding the air clear enough to let me enjoy the view; but I have always been disappointed. Wherever one looks, clouds of smoke rise from the burning bush, and the air is full of smoke and vapour. It is a pity, for under more favourable circumstances the panorama of the whole country up to the distant Majeje hills must be truly magnificent. It is of little use taking photographs now, and an outline sketch gives a very poor idea of the scenery. In one of these excursions I went out of my way to make a personal attempt on the Makonde bush. The present edge of the plateau is the result of a far-reaching process of destruction through erosion and denudation. The Makonde strata are everywhere cut into by ravines, which, though short, are hundreds of yards in depth. In consequence of the loose stratification of these beds, not only are the walls of these ravines nearly vertical, but their upper end is closed by an equally steep escarpment, so that the

western edge of the Makonde plateau is hemmed in by a series of deep, basin-like valleys. In order to get from one side of such a ravine to the other, I cut my way through the bush with a dozen of my men. It was a very open part, with more grass than scrub, but even so the short stretch of less than two hundred yards was very hard work; at the end of it the men's calicoes were in rags and they themselves bleeding from hundreds of scratches, while even our strong khaki suits had not escaped scatheless.



NATIVE PATH THROUGH THE MAKONDE BUSH, NEAR MAHUTA

I see increasing reason to believe that the view formed some time back as to the origin of the Makonde bush is the correct one. I have no doubt that it is not a natural product, but the result of human occupation. Those parts of the high country where man—as a very slight amount of practice enables the eye to perceive at once—has not yet penetrated with axe and hoe, are still occupied by a splendid timber forest quite able to sustain a comparison with our mixed forests in Germany. But wherever man has once built his hut or tilled his field, this horrible bush springs up. Every phase of this process

may be seen in the course of a couple of hours' walk along the main road. From the bush to right or left, one hears the sound of the axe—not from one spot only, but from several directions at once. A few steps further on, we can see what is taking place. The brush has been cut down and piled up in heaps to the height of a yard or more, between which the trunks of the large trees stand up like the last pillars of a magnificent ruined building. These, too, present a melancholy spectacle: the destructive Makonde have ringed them—cut a broad strip of bark all round to ensure their dying off—and also piled up pyramids of brush round them. Father and son, mother and son-in-law, are chopping away perseveringly in the background—too busy, almost, to look round at the white stranger, who usually excites so much interest. If you pass by the same place a week later, the piles of brushwood have disappeared and a thick layer of ashes has taken the place of the green forest. The large trees stretch their smouldering trunks and branches in dumb accusation to heaven—if they have not already fallen and been more or less reduced to ashes, perhaps only showing as a white stripe on the dark ground.

This work of destruction is carried out by the Makonde alike on the virgin forest and on the bush which has sprung up on sites already cultivated and deserted. In the second case they are saved the trouble of burning the large trees, these being entirely absent in the secondary bush.

After burning this piece of forest ground and loosening it with the hoe, the native sows his corn and plants his vegetables. All over the country, he goes in for bed-culture, which requires, and, in fact, receives, the most careful attention. Weeds are nowhere tolerated in the south of German East Africa. The crops may fail on the plains, where droughts are frequent, but never on the plateau with its abundant rains and heavy dews. Its fortunate inhabitants even have the satisfaction of seeing the proud Wayao and Wamakua working for them as labourers, driven by hunger to serve where they were accustomed to rule.

But the light, sandy soil is soon exhausted, and would yield no harvest the second year if cultivated twice running. This fact has been familiar to the native for ages; consequently he provides in time, and, while his crop is growing, prepares the next plot with axe and firebrand. Next year he plants this with his various crops and

lets the first piece lie fallow. For a short time it remains waste and desolate; then nature steps in to repair the destruction wrought by man; a thousand new growths spring out of the exhausted soil, and even the old stumps put forth fresh shoots. Next year the new growth is up to one's knees, and in a few years more it is that terrible, impenetrable bush, which maintains its position till the black occupier of the land has made the round of all the available sites and come back to his starting point.

The Makonde are, body and soul, so to speak, one with this bush. According to my Yao informants, indeed, their name means nothing else but "bush people." Their own tradition says that they have been settled up here for a very long time, but to my surprise they laid great stress on an original immigration. Their old homes were in the south-east, near Mikindani and the mouth of the Rovuma, whence their peaceful forefathers were driven by the continual raids of the Sakalavas from Madagascar and the warlike Shirazis[47] of the coast, to take refuge on the almost inaccessible plateau. I have studied African ethnology for twenty years, but the fact that changes of population in this apparently quiet and peaceable corner of the earth could have been occasioned by outside enterprises taking place on the high seas, was completely new to me. It is, no doubt, however, correct.

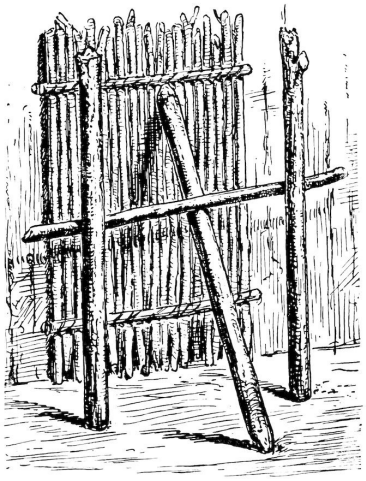
The charming tribal legend of the Makonde—besides informing us of other interesting matters—explains why they have to live in the thickest of the bush and a long way from the edge of the plateau, instead of making their permanent homes beside the purling brooks and springs of the low country.

"The place where the tribe originated is Mahuta, on the southern side of the plateau towards the Rovuma, where of old time there was nothing but thick bush. Out of this bush came a man who never washed himself or shaved his head, and who ate and drank but little. He went out and made a human figure from the wood of a tree growing in the open country, which he took home to his abode in the bush and there set it upright. In the night this image came to life and was a woman. The man and woman went down together to the Rovuma to wash themselves. Here the woman gave birth to a still-born child. They left that place and passed over the high land into the valley of the Mbemkuru, where the woman had another child, which

was also born dead. Then they returned to the high bush country of Mahuta, where the third child was born, which lived and grew up. In course of time, the couple had many more children, and called themselves Wamatanda. These were the ancestral stock of the Makonde, also called Wamakonde,[48] *i.e.*, aborigines. Their forefather, the man from the bush, gave his children the command to bury their dead upright, in memory of the mother of their race who was cut out of wood and awoke to life when standing upright. He also warned them against settling in the valleys and near large streams, for sickness and death dwelt there. They were to make it a rule to have their huts at least an hour's walk from the nearest watering-place; then their children would thrive and escape illness."

The explanation of the name Makonde given by my informants is somewhat different from that contained in the above legend, which I extract from a little book (small, but packed with information), by Pater Adams, entitled *Lindi und sein Hinterland*. Otherwise, my results agree exactly with the statements of the legend. Washing? *Hapana*—there is no such thing. Why should they do so? As it is, the supply of water scarcely suffices for cooking and drinking; other people do not wash, so why should the Makonde distinguish himself by such needless eccentricity? As for shaving the head, the short, woolly crop scarcely needs it,[49] so the second ancestral precept is likewise easy enough to follow. Beyond this, however, there is nothing ridiculous in the ancestor's advice. I have obtained from various local artists a fairly large number of figures carved in wood, ranging from fifteen to twenty-three inches in height, and representing women belonging to the great group of the Mavia, Makonde, and Matambwe tribes. The carving is remarkably well done and renders the female type with great accuracy, especially the keloid ornamentation, to be described later on. As to the object and meaning of their works the sculptors either could or (more probably) would tell me nothing, and I was forced to content myself with the scanty information vouchsafed by one man, who said that the figures were merely intended to represent the *nembo*—the artificial deformations of *pelele*, ear-discs, and keloids. The legend recorded by Pater Adams places these figures in a new light. They must surely be more than mere dolls; and we may even venture to assume that they are—though the majority of present-day Makonde are probably unaware of the fact—representations of the tribal ancestress.

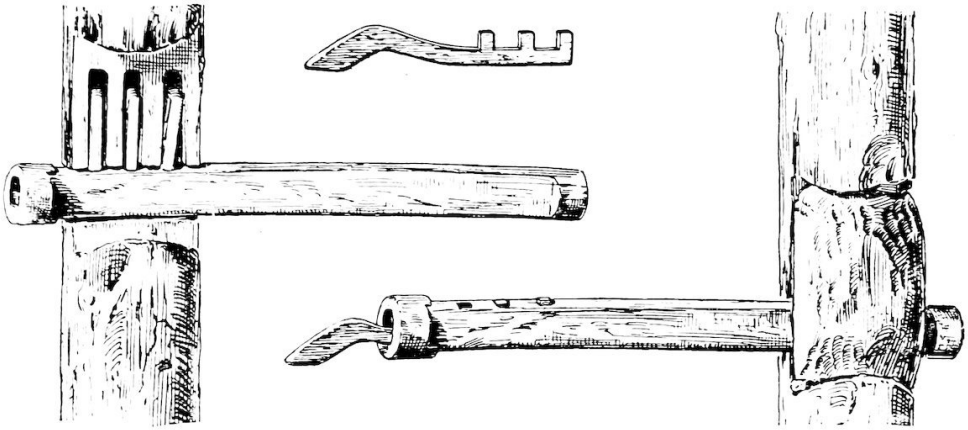
The references in the legend to the descent from Mahuta to the Rovuma, and to a journey across the highlands into the Mbekuru valley, undoubtedly indicate the previous history of the tribe, the travels of the ancestral pair typifying the migrations of their descendants. The descent to the neighbouring Rovuma valley, with its extraordinary fertility and great abundance of game, is intelligible at a glance—but the crossing of the Lukuledi depression, the ascent to the Rondo Plateau and the descent to the Mbemkuru, also lie within the bounds of probability, for all these districts have exactly the same character as the extreme south. Now, however, comes a point of especial interest for our bacteriological age. The primitive Makonde did not enjoy their lives in the marshy river-valleys. Disease raged among them, and many died. It was only after they had returned to their original home near Mahuta, that the health conditions of these people improved. We are very apt to think of the African as a stupid person whose ignorance of nature is only equalled by his fear of it, and who looks on all mishaps as caused by evil spirits and malignant natural powers. It is much more correct to assume in this case that the people very early learnt to distinguish districts infested with malaria from those where it is absent.



USUAL METHOD OF CLOSING HUT-DOOR

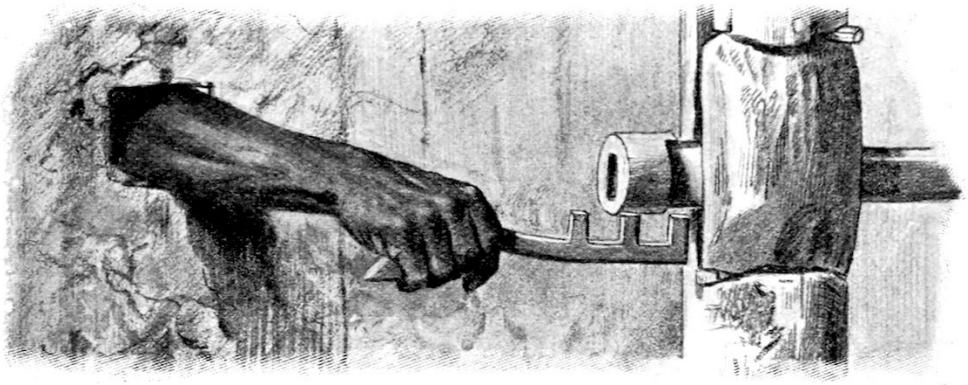
This knowledge is crystallized in the ancestral warning against settling in the valleys and near the great waters, the dwelling-places of disease and death. At the same time, for security against the hostile Mavia south of the Rovuma, it was enacted that every settlement must be not less than a certain distance from the southern edge of the plateau. Such in fact is their mode of life at the present day. It is not such a bad one, and certainly they are both safer and more comfortable than the Makua, the recent intruders from the south, who have made good their footing on the western edge of the plateau, extending over a fairly wide belt of country. Neither Makua nor Makonde show in their dwellings anything of the size and comeliness of the Yao houses in the plain, especially at Masasi, Chingulungulu and Zuza's. Jumbe Chauro, a Makonde hamlet not far from Newala, on the road to Mahuta, is the

most important settlement of the tribe I have yet seen, and has fairly spacious huts. But how slovenly is their construction compared with the palatial residences of the elephant-hunters living in the plain. The roofs are still more untidy than in the general run of huts during the dry season, the walls show here and there the scanty beginnings or the lamentable remains of the mud plastering, and the interior is a veritable dog-kennel; dirt, dust and disorder everywhere. A few huts only show any attempt at division into rooms, and this consists merely of very roughly-made bamboo partitions. In one point alone have I noticed any indication of progress—in the method of fastening the door. Houses all over the south are secured in a simple but ingenious manner. The door consists of a set of stout pieces of wood or bamboo, tied with bark-string to two cross-pieces, and moving in two grooves round one of the door-posts, so as to open inwards. If the owner wishes to leave home, he takes two logs as thick as a man's upper arm and about a yard long. One of these is placed obliquely against the middle of the door from the inside, so as to form an angle of from 60° to 75° with the ground. He then places the second piece horizontally across the first, pressing it downward with all his might. It is kept in place by two strong posts planted in the ground a few inches inside the door. This fastening is absolutely safe, but of course cannot be applied to both doors at once, otherwise how could the owner leave or enter his house? I have not yet succeeded in finding out how the back door is fastened.



MAKONDE LOCK AND KEY AT JUMBE CHAURO

This is the general way of closing a house. The Makonde at Jumbe Chauro, however, have a much more complicated, solid and original one. Here, too, the door is as already described, except that there is only one post on the inside, standing by itself about six inches from one side of the doorway. Opposite this post is a hole in the wall just large enough to admit a man's arm. The door is closed inside by a large wooden bolt passing through a hole in this post and pressing with its free end against the door. The other end has three holes into which fit three pegs running in vertical grooves inside the post. The door is opened with a wooden key about a foot long, somewhat curved and sloped off at the butt; the other end has three pegs corresponding to the holes, in the bolt, so that, when it is thrust through the hole in the wall and inserted into the rectangular opening in the post, the pegs can be lifted and the bolt drawn out.[50]



MODE OF INSERTING THE KEY

With no small pride first one householder and then a second showed me on the spot the action of this greatest invention of the Makonde Highlands. To both with an admiring exclamation of "*Vizuri sana!*" ("Very fine!"). I expressed the wish to take back these marvels with me to Ulaya, to show the Wazungu what clever fellows the Makonde are. Scarcely five minutes after my return to camp at Newala, the two men came up sweating under the weight of two heavy logs which they laid down at my feet, handing over at the same time the keys of the fallen fortress. Arguing, logically enough, that if the key was wanted, the lock would be wanted with it, they had taken their axes and chopped down the posts—as it never occurred to them to dig them out of the ground and so bring them intact. Thus I have

two badly damaged specimens, and the owners, instead of praise, come in for a blowing-up.

The Makua huts in the environs of Newala are especially miserable; their more than slovenly construction reminds one of the temporary erections of the Makua at Hatia's, though the people here have not been concerned in a war. It must therefore be due to congenital idleness, or else to the absence of a powerful chief. Even the *baraza* at Mlipa's, a short hour's walk south-east of Newala, shares in this general neglect. While public buildings in this country are usually looked after more or less carefully, this is in evident danger of being blown over by the first strong easterly gale. The only attractive object in this whole district is the grave of the late chief Mlipa. I visited it in the morning, while the sun was still trying with partial success to break through the rolling mists, and the circular grove of tall euphorbias, which, with a broken pot, is all that marks the old king's resting-place, impressed one with a touch of pathos. Even my very materially-minded carriers seemed to feel something of the sort, for instead of their usual ribald songs, they chanted solemnly, as we marched on through the dense green of the Makonde bush:—



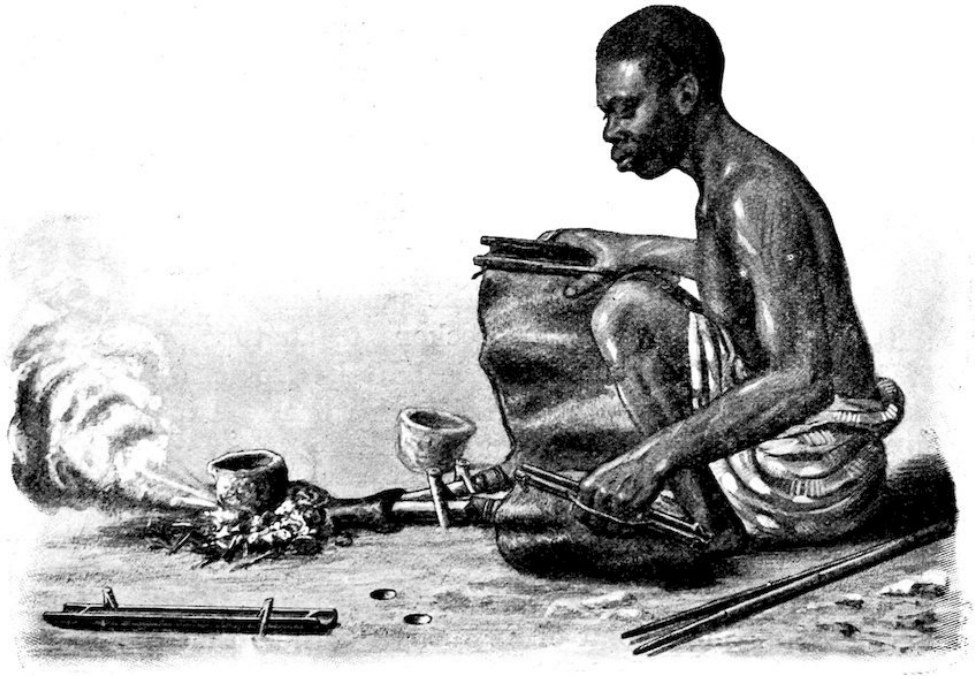
“We shall arrive with the great master; we stand in a row and have no fear about getting our food and our money from the Serkali (the Government). We are not afraid; we are going along with the great master, the lion; we are going down to the coast and back.”

With regard to the characteristic features of the various tribes here on the western edge of the plateau, I can arrive at no other conclusion than the one already come to in the plain, viz., that it is impossible for anyone but a trained anthropologist to assign any given individual at once to his proper tribe. In fact, I think that even an anthropological specialist, after the most careful examination, might find it a difficult task to decide. The whole congeries of peoples collected in the region bounded on the west by the great Central African rift, Tanganyika and Nyasa, and on the east by the Indian Ocean, are closely related to each other—some of their languages are only distinguished from one another as dialects of the same speech, and no doubt all the tribes present the same shape of skull and structure of skeleton. Thus, surely, there can be no very striking differences in outward appearance.



THE ANCESTRESS OF THE MAKONDE

Even did such exist, I should have no time to concern myself with them, for day after day, I have to see or hear, as the case may be—in any case to grasp and record—an extraordinary number of ethnographic phenomena. I am almost disposed to think it fortunate that some departments of inquiry, at least, are barred by external circumstances. Chief among these is the subject of iron-working. We are apt to think of Africa as a country where iron ore is everywhere, so to speak, to be picked up by the roadside, and where it would be quite surprising if the inhabitants had not learnt to smelt the material ready to their hand. In fact, the knowledge of this art ranges all over the continent, from the Kabyles in the north to the Kafirs in the south. Here between the Rovuma and the Lukuledi the conditions are not so favourable. According to the statements of the Makonde, neither ironstone nor any other form of iron ore is known to them. They have not therefore advanced to the art of smelting the metal, but have hitherto bought all their iron implements from neighbouring tribes. Even in the plain the inhabitants are not much better off. Only one man now living is said to understand the art of smelting iron. This old *fundi* lives close to Huwe, that isolated, steep-sided block of granite which rises out of the green solitude between Masasi and Chingulungulu, and whose jagged and splintered top meets the traveller's eye everywhere. While still at Masasi I wished to see this man at work, but was told that, frightened by the rising, he had retired across the Rovuma, though he would soon return. All subsequent inquiries as to whether the *fundi* had come back met with the genuine African answer, "*Bado*" ("Not yet").

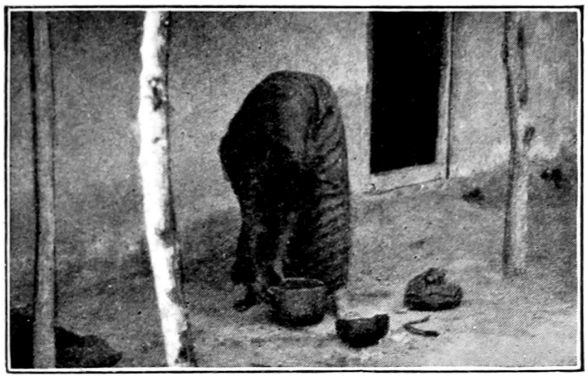


BRAZIER

Some consolation was afforded me by a brassfounder, whom I came across in the bush near Akundonde's. This man is the favourite of women, and therefore no doubt of the gods; he welds the glittering brass rods purchased at the coast into those massive, heavy rings which, on the wrists and ankles of the local fair ones, continually give me fresh food for admiration. Like every decent master-craftsman he had all his tools with him, consisting of a pair of bellows, three crucibles and a hammer—nothing more, apparently. He was quite willing to show his skill, and in a twinkling had fixed his bellows on the ground. They are simply two goat-skins, taken off whole, the four legs being closed by knots, while the upper opening, intended to admit the air, is kept stretched by two pieces of wood. At the lower end of the skin a smaller opening is left into which a wooden tube is stuck. The *fundi* has quickly borrowed a heap of wood-embers from the nearest hut; he then fixes the free ends of the two tubes into an earthen pipe, and clamps them to the ground by means of a bent piece of wood. Now he fills one of his small clay crucibles, the dross on which shows that they have been long in use, with the yellow material, places it in the midst of the embers, which, at present are only faintly glimmering, and begins his work. In quick alternation

the smith's two hands move up and down with the open ends of the bellows; as he raises his hand he holds the slit wide open, so as to let the air enter the skin bag unhindered. In pressing it down he closes the bag, and the air puffs through the bamboo tube and clay pipe into the fire, which quickly burns up. The smith, however, does not keep on with this work, but beckons to another man, who relieves him at the bellows, while he takes some more tools out of a large skin pouch carried on his back. I look on in wonder as, with a smooth round stick about the thickness of a finger, he bores a few vertical holes into the clean sand of the soil. This should not be difficult, yet the man seems to be taking great pains over it. Then he fastens down to the ground, with a couple of wooden clamps, a neat little trough made by splitting a joint of bamboo in half, so that the ends are closed by the two knots. At last the yellow metal has attained the right consistency, and the *fundi* lifts the crucible from the fire by means of two sticks split at the end to serve as tongs. A short swift turn to the left—a tilting of the crucible—and the molten brass, hissing and giving forth clouds of smoke, flows first into the bamboo mould and then into the holes in the ground.

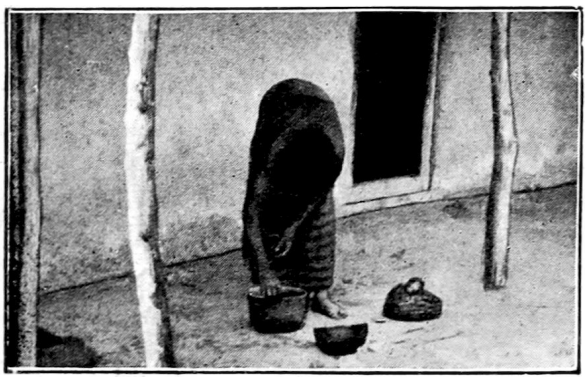
The technique of this backwoods craftsman may not be very far advanced, but it cannot be denied that he knows how to obtain an adequate result by the simplest means. The ladies of highest rank in this country—that is to say, those who can afford it, wear two kinds of these massive brass rings, one cylindrical, the other semicircular in section. The latter are cast in the most ingenious way in the bamboo mould, the former in the circular hole in the sand. It is quite a simple matter for the *fundi* to fit these bars to the limbs of his fair customers; with a few light strokes of his hammer he bends the pliable brass round arm or ankle without further inconvenience to the wearer.



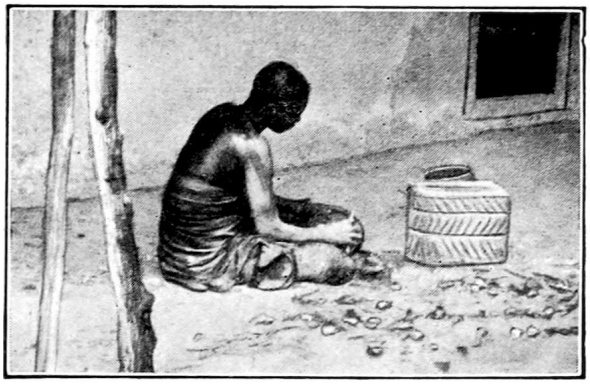
SHAPING THE POT



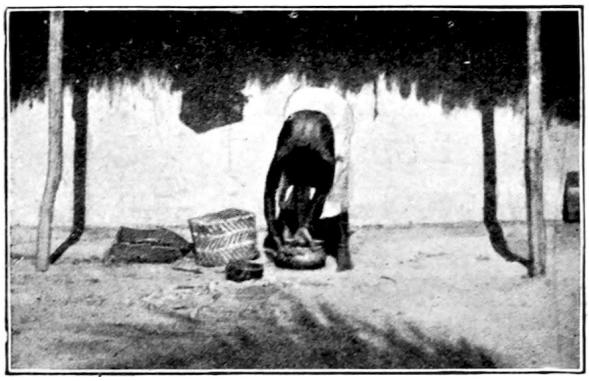
SMOOTHING WITH MAIZE-COB



CUTTING THE EDGE



FINISHING THE BOTTOM



LAST SMOOTHING BEFORE
BURNING



FIRING THE BRUSH-PILE



LIGHTING THE FARTHER SIDE OF THE PILE



TURNING THE RED-HOT VESSEL

NYASA WOMAN MAKING POTS AT MASASI

Pottery is an art which must always and everywhere excite the interest of the student, just because it is so intimately connected with the development of human culture, and because its relics are one of the principal factors in the reconstruction of our own condition in prehistoric times. I shall always remember with pleasure the two or three afternoons at Masasi when Salim Matola's mother, a slightly-built, graceful, pleasant-looking woman, explained to me with touching patience, by means of concrete illustrations, the ceramic art of her people. The only implements for this primitive process were a lump of clay in her left hand, and in the right a calabash containing the following valuables: the fragment of a maize-cob stripped of all its grains, a smooth, oval pebble, about the size of a pigeon's egg, a few chips of gourd-shell, a bamboo splinter about the length of one's hand, a small shell, and a bunch of some herb resembling spinach.



MAKUA WOMAN MAKING A POT. SHOWS THE BEGINNINGS OF THE POTTER'S WHEEL

Nothing more. The woman scraped with the shell a round, shallow hole in the soft, fine sand of the soil, and, when an active young girl had filled the calabash with water for her, she began to knead the clay. As if by magic it gradually assumed the shape of a rough but already well-shaped vessel, which only wanted a little touching up with the instruments before mentioned. I looked out with the closest attention for any indication of the use of the potter's wheel, in however rudimentary a form, but no—*hapana* (there is none). The embryo pot stood firmly in its little depression, and the woman walked round it in a stooping posture, whether she was removing small stones or similar foreign bodies with the maize-cob, smoothing the inner or outer surface with the splinter of bamboo, or later, after letting it dry for a day, pricking in the ornamentation with a pointed bit of gourd-shell, or working out the bottom, or cutting the edge with a sharp bamboo knife, or giving the last touches to the finished vessel. This occupation of the women is infinitely toilsome, but it is without doubt an accurate reproduction of the process in use among our ancestors of the Neolithic and Bronze ages.

There is no doubt that the invention of pottery, an item in human progress whose importance cannot be over-estimated, is due to women. Rough, coarse and unfeeling, the men of the horde range over the countryside. When the united cunning of the hunters has succeeded in killing the game; not one of them thinks of carrying home the spoil. A bright fire, kindled by a vigorous wielding of the drill, is crackling beside them; the animal has been cleaned and cut up *secundum artem*, and, after a slight singeing, will soon disappear under their sharp teeth; no one all this time giving a single thought to wife or child.

To what shifts, on the other hand, the primitive wife, and still more the primitive mother, was put! Not even prehistoric stomachs could endure an unvarying diet of raw food. Something or other suggested the beneficial effect of hot water on the majority of approved but indigestible dishes. Perhaps a neighbour had tried holding the hard

roots or tubers over the fire in a calabash filled with water—or maybe an ostrich-egg-shell, or a hastily improvised vessel of bark. They became much softer and more palatable than they had previously been; but, unfortunately, the vessel could not stand the fire and got charred on the outside. That can be remedied, thought our ancestress, and plastered a layer of wet clay round a similar vessel. This is an improvement; the cooking utensil remains uninjured, but the heat of the fire has shrunk it, so that it is loose in its shell. The next step is to detach it, so, with a firm grip and a jerk, shell and kernel are separated, and pottery is invented. Perhaps, however, the discovery which led to an intelligent use of the burnt-clay shell, was made in a slightly different way. Ostrich-eggs and calabashes are not to be found in every part of the world, but everywhere mankind has arrived at the art of making baskets out of pliant materials, such as bark, bast, strips of palm-leaf, supple twigs, etc. Our inventor has no water-tight vessel provided by nature. "Never mind, let us line the basket with clay." This answers the purpose, but alas! the basket gets burnt over the blazing fire, the woman watches the process of cooking with increasing uneasiness, fearing a leak, but no leak appears. The food, done to a turn, is eaten with peculiar relish; and the cooking-vessel is examined, half in curiosity, half in satisfaction at the result. The plastic clay is now hard as stone, and at the same time looks exceedingly well, for the neat plaiting of the burnt basket is traced all over it in a pretty pattern. Thus, simultaneously with pottery, its ornamentation was invented.

Primitive woman has another claim to respect. It was the man, roving abroad, who invented the art of producing fire at will, but the woman, unable to imitate him in this, has been a Vestal from the earliest times. Nothing gives so much trouble as the keeping alight of the smouldering brand, and, above all, when all the men are absent from the camp. Heavy rain-clouds gather, already the first large drops are falling, the first gusts of the storm rage over the plain. The little flame, a greater anxiety to the woman than her own children, flickers unsteadily in the blast. What is to be done? A sudden thought occurs to her, and in an instant she has constructed a primitive hut out of strips of bark, to protect the flame against rain and wind.

This, or something very like it, was the way in which the principle of the house was discovered; and even the most hardened misogynist

cannot fairly refuse a woman the credit of it. The protection of the hearth-fire from the weather is the germ from which the human dwelling was evolved. Men had little, if any share, in this forward step, and that only at a late stage. Even at the present day, the plastering of the housewall with clay and the manufacture of pottery are exclusively the women's business. These are two very significant survivals. Our European kitchen-garden, too, is originally a woman's invention, and the hoe, the primitive instrument of agriculture, is, characteristically enough, still used in this department. But the noblest achievement which we owe to the other sex is unquestionably the art of cookery. Roasting alone—the oldest process—is one for which men took the hint (a very obvious one) from nature. It must have been suggested by the scorched carcase of some animal overtaken by the destructive forest-fires. But boiling—the process of improving organic substances by the help of water heated to boiling-point—is a much later discovery. It is so recent that it has not even yet penetrated to all parts of the world. The Polynesians understand how to steam food, that is, to cook it, neatly wrapped in leaves, in a hole in the earth between hot stones, the air being excluded, and (sometimes) a few drops of water sprinkled on the stones; but they do not understand boiling.

To come back from this digression, we find that the slender Nyasa woman has, after once more carefully examining the finished pot, put it aside in the shade to dry. On the following day she sends me word by her son, Salim Matola, who is always on hand, that she is going to do the burning, and, on coming out of my house, I find her already hard at work. She has spread on the ground a layer of very dry sticks, about as thick as one's thumb, has laid the pot (now of a yellowish-grey colour) on them, and is piling brushwood round it. My faithful Pesa mbili, the *mnyampara*, who has been standing by, most obligingly, with a lighted stick, now hands it to her. Both of them, blowing steadily, light the pile on the lee side, and, when the flame begins to catch, on the weather side also. Soon the whole is in a blaze, but the dry fuel is quickly consumed and the fire dies down, so that we see the red-hot vessel rising from the ashes. The woman turns it continually with a long stick, sometimes one way and sometimes another, so that it may be evenly heated all over. In twenty minutes she rolls it out of the ash-heap, takes up the bundle of spinach, which has been lying for two days in a jar of water, and

sprinkles the red-hot clay with it. The places where the drops fall are marked by black spots on the uniform reddish-brown surface. With a sigh of relief, and with visible satisfaction, the woman rises to an erect position; she is standing just in a line between me and the fire, from which a cloud of smoke is just rising: I press the ball of my camera, the shutter clicks—the apotheosis is achieved! Like a priestess, representative of her inventive sex, the graceful woman stands: at her feet the hearth-fire she has given us beside her the invention she has devised for us, in the background the home she has built for us.

At Newala, also, I have had the manufacture of pottery carried on in my presence. Technically the process is better than that already described, for here we find the beginnings of the potter's wheel, which does not seem to exist in the plains; at least I have seen nothing of the sort. The artist, a frightfully stupid Makua woman, did not make a depression in the ground to receive the pot she was about to shape, but used instead a large potsherd. Otherwise, she went to work in much the same way as Salim's mother, except that she saved herself the trouble of walking round and round her work by squatting at her ease and letting the pot and potsherd rotate round her; this is surely the first step towards a machine. But it does not follow that the pot was improved by the process. It is true that it was beautifully rounded and presented a very creditable appearance when finished, but the numerous large and small vessels which I have seen, and, in part, collected, in the "less advanced" districts, are no less so. We moderns imagine that instruments of precision are necessary to produce excellent results. Go to the prehistoric collections of our museums and look at the pots, urns and bowls of our ancestors in the dim ages of the past, and you will at once perceive your error.



MAKING LONGITUDINAL CUT IN BARK



DRAWING THE BARK OFF THE LOG



REMOVING THE OUTER BARK



BEATING THE BARK



WORKING THE BARK-CLOTH AFTER BEATING, TO MAKE IT SOFT

MANUFACTURE OF BARK-CLOTH AT NEWALA

To-day, nearly the whole population of German East Africa is clothed in imported calico. This was not always the case; even now in some parts of the north dressed skins are still the prevailing wear, and in the north-western districts—east and north of Lake Tanganyika—lies a zone where bark-cloth has not yet been superseded. Probably not many generations have passed since such bark fabrics and kilts of skins were the only clothing even in the south. Even to-day, large quantities of this bright-red or drab material are still to be found; but if we wish to see it, we must look in the granaries and on the drying stages inside the native huts, where

it serves less ambitious uses as wrappings for those seeds and fruits which require to be packed with special care. The salt produced at Masasi, too, is packed for transport to a distance in large sheets of bark-cloth. Wherever I found it in any degree possible, I studied the process of making this cloth. The native requisitioned for the purpose arrived, carrying a log between two and three yards long and as thick as his thigh, and nothing else except a curiously-shaped mallet and the usual long, sharp and pointed knife which all men and boys wear in a belt at their backs without a sheath—*horribile dictu!* [51] Silently he squats down before me, and with two rapid cuts has drawn a couple of circles round the log some two yards apart, and slits the bark lengthwise between them with the point of his knife. With evident care, he then scrapes off the outer rind all round the log, so that in a quarter of an hour the inner red layer of the bark shows up brightly-coloured between the two untouched ends. With some trouble and much caution, he now loosens the bark at one end, and opens the cylinder. He then stands up, takes hold of the free edge with both hands, and turning it inside out, slowly but steadily pulls it off in one piece. Now comes the troublesome work of scraping all superfluous particles of outer bark from the outside of the long, narrow piece of material, while the inner side is carefully scrutinised for defective spots. At last it is ready for beating. Having signalled to a friend, who immediately places a bowl of water beside him, the artificer damps his sheet of bark all over, seizes his mallet, lays one end of the stuff on the smoothest spot of the log, and hammers away slowly but continuously. “Very simple!” I think to myself. “Why, I could do that, too!”—but I am forced to change my opinions a little later on; for the beating is quite an art, if the fabric is not to be beaten to pieces. To prevent the breaking of the fibres, the stuff is several times folded across, so as to interpose several thicknesses between the mallet and the block. At last the required state is reached, and the *fundi* seizes the sheet, still folded, by both ends, and wrings it out, or calls an assistant to take one end while he holds the other. The cloth produced in this way is not nearly so fine and uniform in texture as the famous Uganda bark-cloth, but it is quite soft, and, above all, cheap.

Now, too, I examine the mallet. My craftsman has been using the simpler but better form of this implement, a conical block of some hard wood, its base—the striking surface—being scored across and

across with more or less deeply-cut grooves, and the handle stuck into a hole in the middle. The other and earlier form of mallet is shaped in the same way, but the head is fastened by an ingenious network of bark strips into the split bamboo serving as a handle. The observation so often made, that ancient customs persist longest in connection with religious ceremonies and in the life of children, here finds confirmation. As we shall soon see, bark-cloth is still worn during the *unyago*,[52] having been prepared with special solemn ceremonies; and many a mother, if she has no other garment handy, will still put her little one into a kilt of bark-cloth, which, after all, looks better, besides being more in keeping with its African surroundings, than the ridiculous bit of print from Ulaya.



MAKUA WOMEN